



MMA093 Rigid Body Dynamics

Assignment 1

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Part 1: The latch design problem

In Latch design problem the build model gives the Measure on the Spring force which is needed for a measurement of the clamping force when the latch is clamped. The spring force measure represents the clamping force. Refer fig 1.

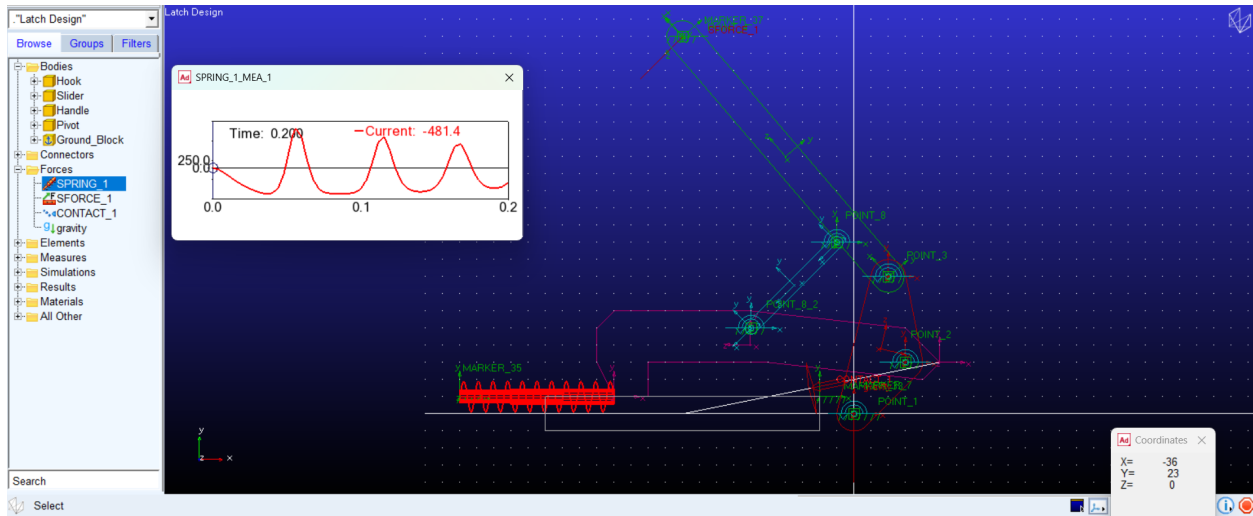


Figure 1: Measure on spring force

Angle measure is created to ensure that the handle pushes down far enough to engage the clutch. Refer fig 2.

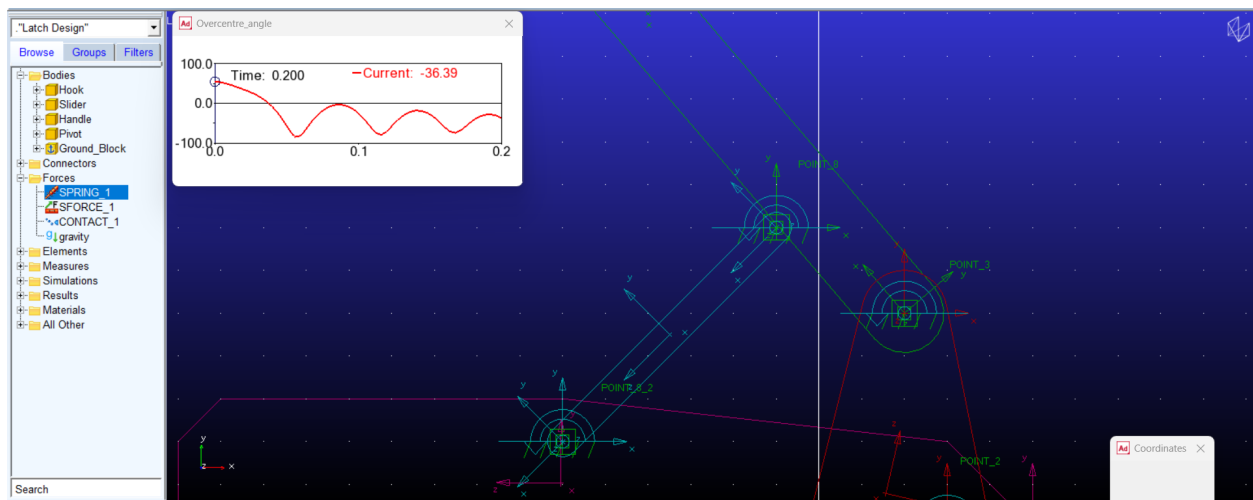


Figure 2: Angle Measure

In test model as the simulation proceeds the strip charts is updated. The strip charts for the spring force and angle measure to show that the sensor stopped the simulation. Refer fig 3.

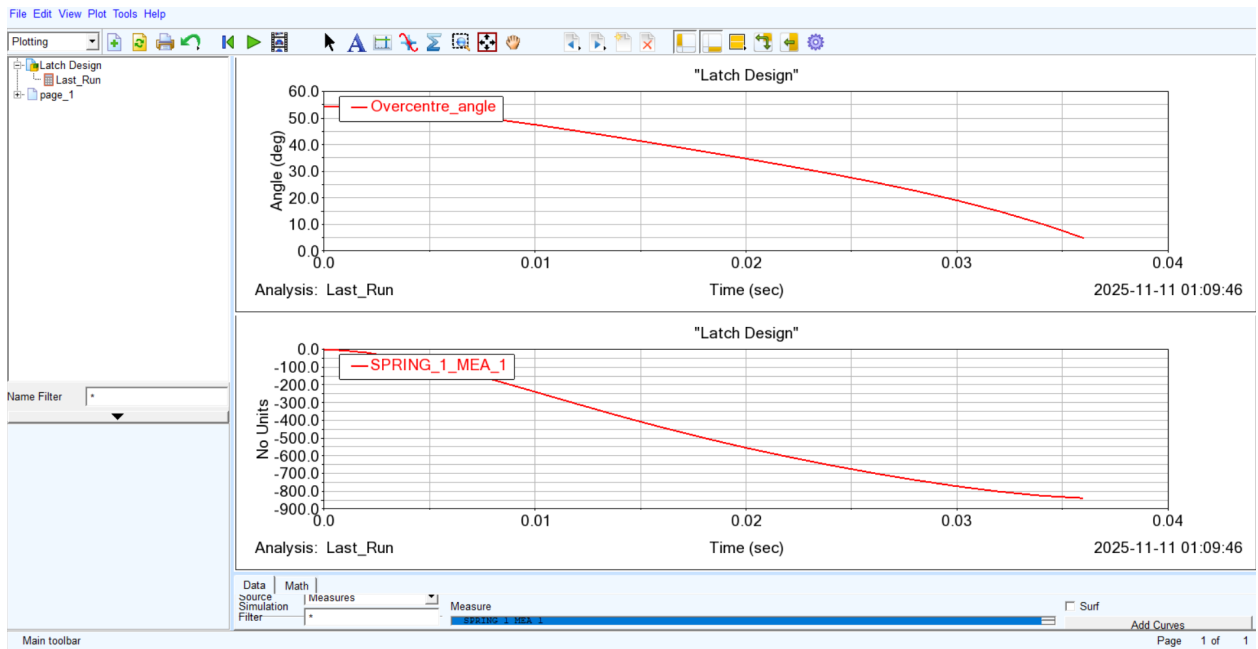
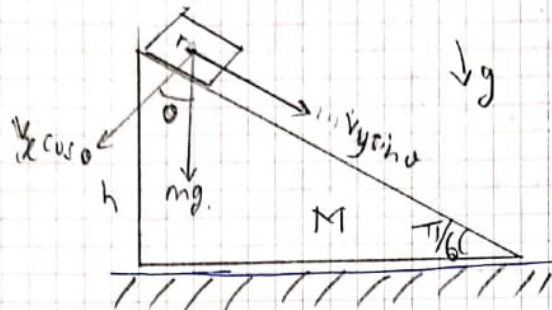


Figure 3: Overcentre angle and Spring force

Part 2: The box on a wedge

2. Derive an expression of velocity.



m = mass of box M = Mass of wedge.
 v = Speed of box relative to wedge.
 U = Speed of wedge (horizontal velocity).
 No friction $\mu_1 = \mu_2 = 0$

Box absolute horizontal velocity.

$$V_x = v \cos \theta + U. \quad \text{--- (1)}$$

Box absolute vertical velocity

$$V_y = -v \sin \theta. \quad \text{--- (2)}$$

Conservation of horizontal momentum

$$mV_x + MU = 0. \quad \text{--- (3)}$$

$$m(v \cos \theta + U) + MU = 0. \quad \text{(Substitute (1) in (3))}$$

$$mV \cos \theta + mU + MU = 0.$$

$$mV \cos \theta + U(m + M) = 0.$$

$$\boxed{U_{\text{wedge}} = -\frac{mV \cos \theta}{m + M}} \quad \text{--- (4)}$$

$$\begin{aligned}
 U_{\text{wedge}} &= -\frac{1 \times 3.348 \times \cos 30^\circ}{1 + 5} \\
 \boxed{U_{\text{wedge}} = -0.486 \text{ m/s}}
 \end{aligned}$$

⑥ Kinetic Energy Conservation

$$mgh = \frac{1}{2} mV^2 + \frac{1}{2} MU^2 \quad \text{--- (5)}$$

$$mgh = \frac{1}{2} m \left[v \cos^2 (U + V \cos \theta) + (V \sin \theta)^2 \right] + \frac{1}{2} MU^2 \quad \text{--- (6)}$$

$$\begin{aligned}
 (U + V \cos \theta)^2 + (V \sin \theta)^2 &= U^2 + V^2 \cos^2 \theta + 2UV \cos \theta + V^2 \sin^2 \theta \\
 &= U^2 + 2UV \cos \theta + V^2 (\cos^2 \theta + \sin^2 \theta) \\
 &= U^2 + 2UV \cos \theta + V^2 - \textcircled{7} [\cos^2 \theta + \sin^2 \theta = 1].
 \end{aligned}$$

Substitute $\textcircled{7}$ in $\textcircled{6}$.

$$\begin{aligned}
 &= \frac{1}{2} m [U^2 + 2UV \cos \theta + V^2] + \frac{1}{2} M U^2 \\
 &= \frac{1}{2} m U^2 + mUV \cos \theta + \frac{1}{2} m V^2 + \frac{1}{2} M U^2 \\
 &= \cancel{\frac{1}{2} m U^2} + \frac{1}{2} (m+M) U^2 + mUV \cos \theta + \frac{1}{2} m V^2 - \textcircled{8}
 \end{aligned}$$

Substitute $\textcircled{4}$ in $\textcircled{8}$.

$$mgh = \frac{1}{2} (m+M) \left(\frac{-mV \cos \theta}{m+M} \right)^2 + m \left(\frac{-mV \cos \theta}{m+M} \right) V \cos \theta + \frac{1}{2} m V^2.$$

$$a = \frac{mV \cos \theta}{m+M} \Rightarrow V = -aV$$

$$mgh = \frac{1}{2} (m+M) (-aV)^2 + m (-aV) V \cos \theta + \frac{1}{2} m V^2.$$

$$\frac{2mgh}{(m+M)a^2} = -maV \cos \theta + \frac{1}{2} m V^2$$

$$= \frac{2mgh}{(m+M) \frac{m^2 V^2 \cos^2 \theta}{(m+M)^2}} = \frac{-m^2 V^2 \cos^2 \theta}{(m+M)} + \frac{1}{2} m V^2$$

$$\begin{aligned}
 mgh &= \frac{1}{2} (m+M) a^2 - maV \cos \theta + \frac{1}{2} m V^2 \\
 mgh &= a \left[\frac{1}{2} (m+M) a - mV \cos \theta \right] + \frac{1}{2} m V^2
 \end{aligned}$$

$$\frac{mgh}{a} = \frac{1}{2} (m+M) a - m \cancel{a \cos \theta} + \frac{1}{2} m v^2.$$

$$mgh = \frac{1}{2} (m+M) a^2 v^2 - m a v^2 \cos \theta + \frac{1}{2} m v^2. \quad (9)$$

$$\text{or } \frac{1}{2} (m+M) a^2 v^2 = \frac{1}{2} (m+M) \frac{m^2 \cos^2 \theta}{(m+M)^2} v^2 = \frac{1}{2} \frac{m^2 \cos^2 \theta}{(m+M)} v^2. \quad (10)$$

$$\ddot{x} = m a v^2 \cos \theta = -m \frac{m \cos \theta}{(m+M)} v^2 \cos \theta = - \frac{m^2 \cos^2 \theta}{(m+M)} v^2. \quad (11)$$

Substitute (10), (11) in (9).

$$mgh = \frac{1}{2} \frac{m^2 \cos^2 \theta}{(m+M)} v^2 - \frac{m^2 \cos^2 \theta}{(m+M)} v^2 + \frac{1}{2} m v^2.$$

$$mgh = - \frac{1}{2} \frac{m^2 \cos^2 \theta}{(m+M)} v^2 + \frac{1}{2} m v^2.$$

$$mgh = \frac{1}{2} m v^2 \left[1 - \frac{m \cos^2 \theta}{(m+M)} \right] \Rightarrow \frac{1}{2} m v^2 \left[\frac{(m+M) - m \cos^2 \theta}{(m+M)} \right]$$

$$mgh = \frac{1}{2} m v^2 \left[\frac{M + m(1 - \cos^2 \theta)}{(m+M)} \right] \Rightarrow \frac{1}{2} m v^2 \left[\frac{M + m \sin^2 \theta}{(m+M)} \right]$$

$$\frac{2mgh}{M + m \sin^2 \theta} \Rightarrow v^2 \Rightarrow v = \frac{2gh(m+M)}{M + m \sin^2 \theta}$$

$$v_{\text{relative}} = \sqrt{\frac{2gh(m+M)}{M + m \sin^2 \theta}}$$

$$V_{\text{box}} = \sqrt{(V_w + V_{\text{rel}} \cos \theta)^2 + V_r^2}$$

$$v = \sqrt{\frac{2gh(m+M)}{M(1+m\sin^2\theta)}} =$$

$$m = 1 \text{ kg} \quad M = 5 \text{ kg}$$

$$h = 0.5 \text{ m.} \quad \theta = \pi/6$$

$$\text{deg} = \text{rad} \times \frac{180}{\pi}$$

$$= \frac{\pi}{6} \times \frac{180}{\pi} = 30^\circ$$

$$v = \sqrt{\frac{2 \times 9.81 \times 0.50 (1+5)}{5 + 1 \times \sin^2(30)}}.$$

$$v = 3.3483 \text{ m/s}$$

Relative to

Velocity of the box.

$$V_b = \sqrt{(V_w + V_{rel} \cos \theta)^2 + (V_{rel} \sin \theta)^2}$$

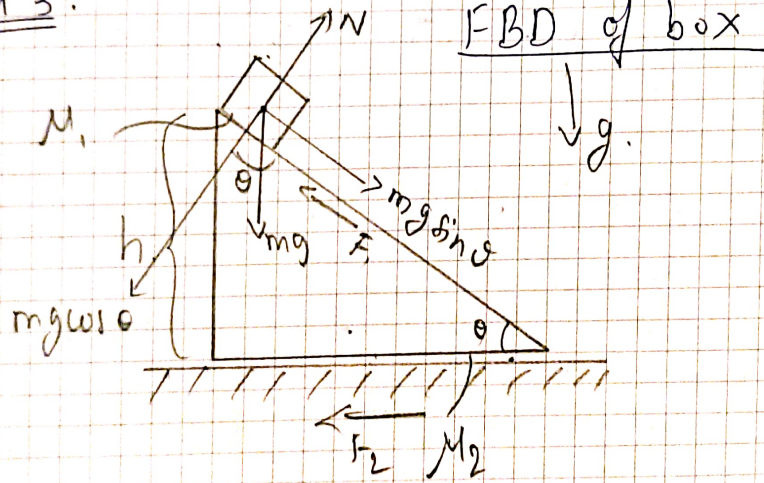
$$= \sqrt{(-0.48 + 3.348 \cos 30)^2 + (3.348 \sin 30)^2}$$

$$= 2.9421$$

$$V_b = 2.9421 \text{ m/s}$$

3. The value of μ_2

Problem 3.



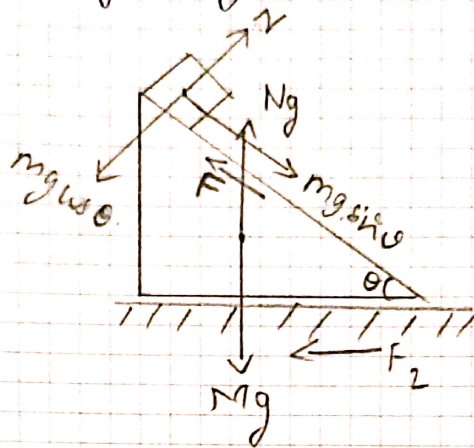
Σ Forces acting parallel on box

$$F - mg \sin \theta = 0 \Rightarrow ma = mg \sin \theta \Rightarrow a = \sin \theta \quad (1)$$

Σ Forces acting perpendicular on box

$$mg \cos \theta - N = 0$$
$$\Rightarrow N = mg \cos \theta \quad (2)$$

Free FBD of Wedge



Σ Force parallel acting parallel on wedge

$$-F_2 + N \sin \theta = 0 \Rightarrow F_2 = N \sin \theta \quad \left[\begin{array}{l} \text{from (1).} \\ N = mg \cos \theta \end{array} \right]$$

$$F_2 = mg \cos \theta \sin \theta \quad \text{--- (3)}$$

Σ Force acting perpendicular on wedge

$$Mg - N_g + N \cos \theta = 0 \Rightarrow N_g = Mg + mg \cos^2 \theta \quad \text{--- (4)}$$

Static Equilibrium.

$$F_2 = \mu_2 N_g \Rightarrow \mu_2 = \frac{F_2}{N_g}$$

$$\mu_2 = \frac{mg \cos \theta \sin \theta}{Mg + mg \cos^2 \theta} \Rightarrow \frac{mg \cos \theta \sin \theta}{g(M + m \cos^2 \theta)}$$

$$\boxed{\mu_2 = \frac{m \cos \theta \sin \theta}{M + m \cos^2 \theta}}$$

$$\mu_2 = \frac{1 \times \cos\left(\frac{\pi}{6}\right) \sin\left(\frac{\pi}{6}\right)}{5 + 1 \times \cos\left(\frac{\pi}{6}\right)^2} = \underline{\underline{0.0753}}$$

4.Refer to Adams file

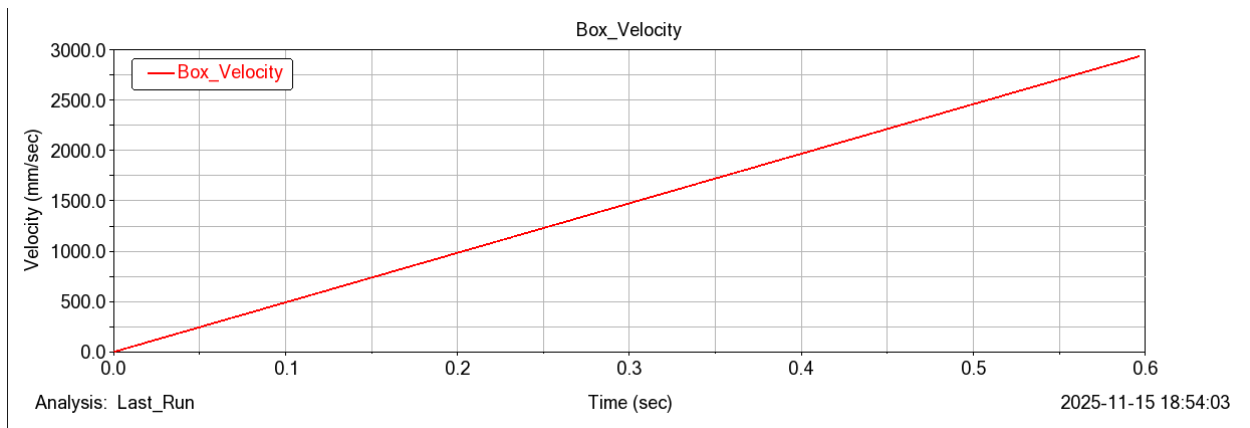


Figure 4: Box velocity(mm/sec) vs Time(sec)

The Box velocity 2.9421(m/s) obtained in the analytical is similar to Adams results.

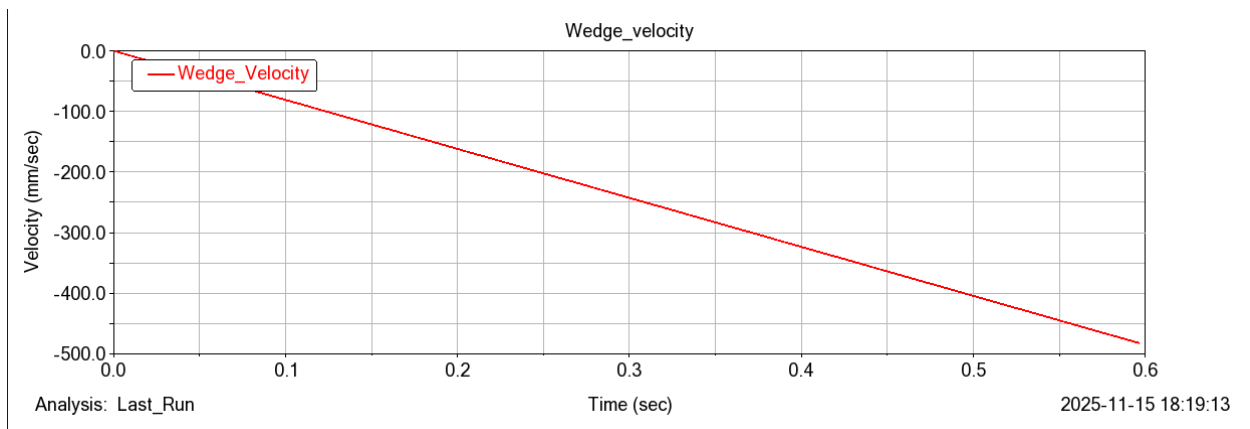


Figure 5: Wedge velocity(mm/sec) vs Time(sec)

The Wedge velocity -0.483(m/s) obtained in the analytical is similar to Adams results.

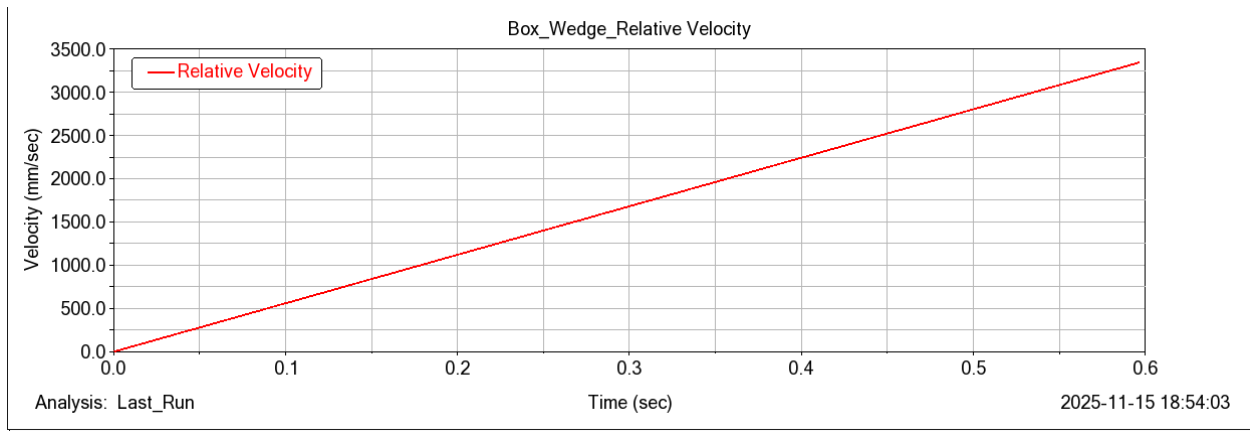


Figure 6: Relative velocity(mm/sec) vs Time(sec)

The Relative velocity 3.3483(m/s) obtained in the analytical is similar to Adams results.

5.Refer to Adams file

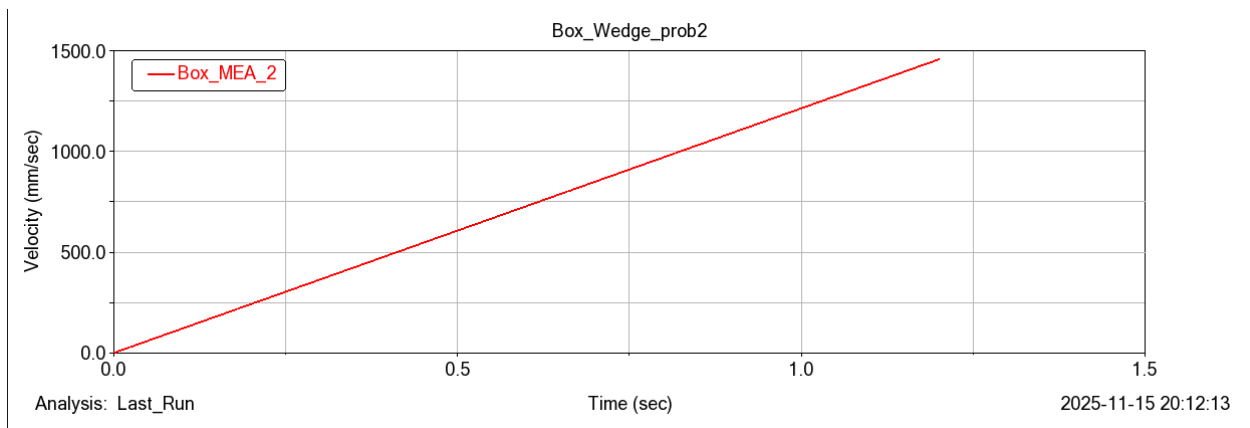


Figure 7: Box velocity(mm/sec) vs Time(sec)

When the friction coefficient between the wedge and the box is $\mu_1 = 0.44$, the velocity of the box is 1.4710(m/s) at end of sliding.

6.Refer to Adams file

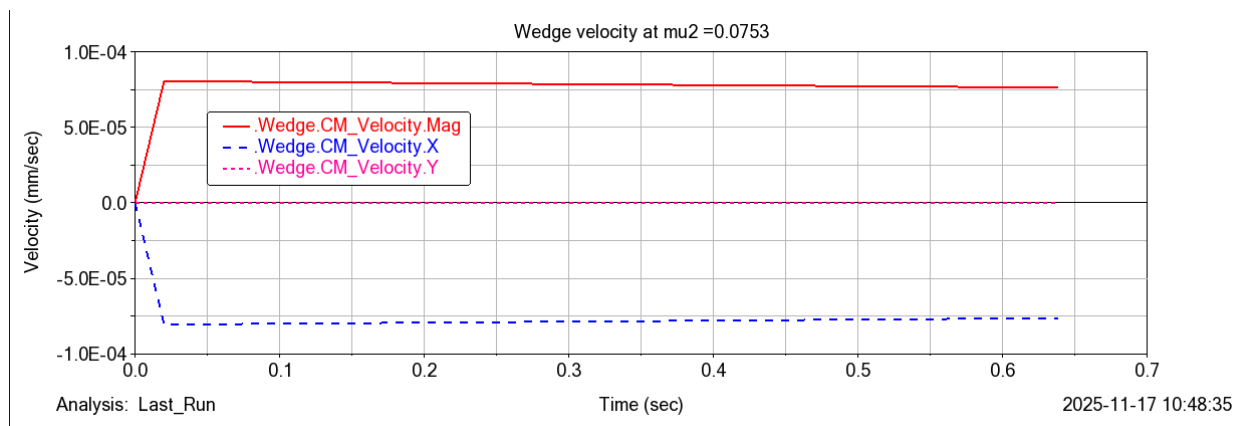


Figure 8: Wedge velocity(mm/sec) vs Time(sec)

When the friction coefficient between the ground and the wedge is $\mu_2 = 0.0753$, the wedge remains at rest where the velocity at X is $-7.0\text{E-}04$ mm/s and velocity at Y is $7.0\text{E-}04$ mm/s which is very small almost equal to zero.

7.Refer to Adams file

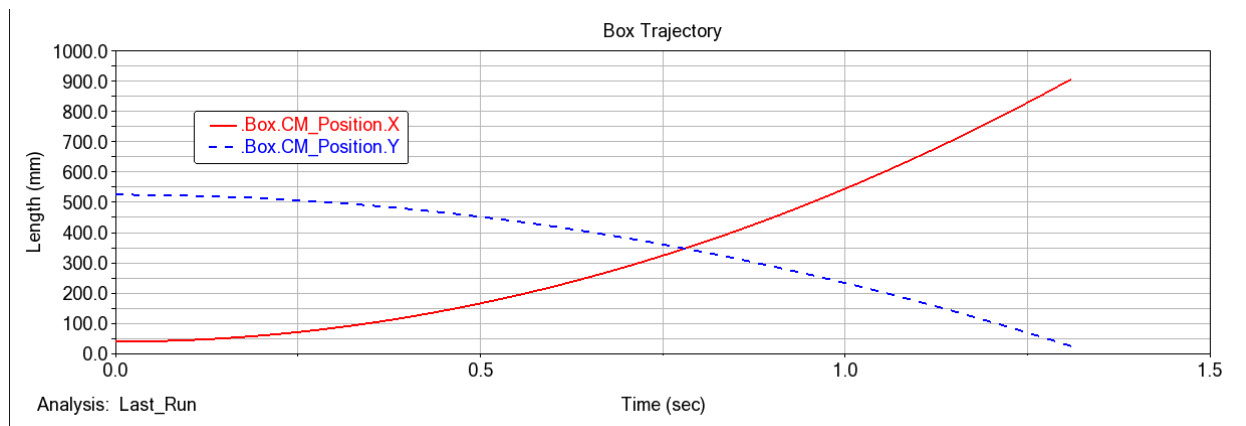


Figure 9: Box Length(mm) vs Time(sec)

At friction coefficient $\mu_1 = 0.44$ and $\mu_2 = 0.0753$, the trajectory of the box is shown above in fig 9.

8. Expression for the constraint equation

Problem 8

2 rigid bodies, each have 3 DOF, x-translation, y-translation & θ -rotation.

Box - 3 DOF

Wedge - 3 DOF

Total unconstrained DOF = 6.

Constraints in the box - wedge problem

Wedge - Translation Joint - only allows horizontal translation $\Rightarrow$ Constraint introduced by this joint is 2

Box - Translation Joint - only allows to travel along the slope of wedge $\rightarrow$ Constraint introduced is 2.

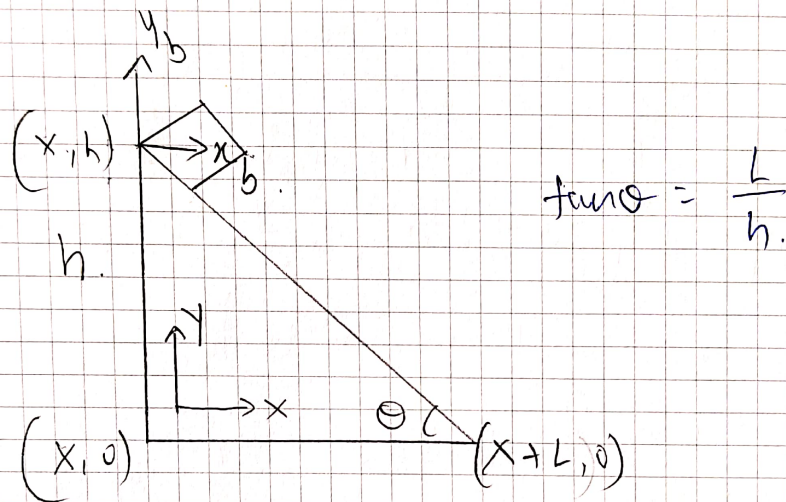
Independent generalised coordinate

Wedge Constraint = 2

Box Constraint = 2

Remaining DOF = $6 - 4 = 2$

2 generalised coordinates needed to describe system configuration.



$$x_b = X + s \cos \theta \quad y_b = h - s \sin \theta \quad - (1)$$

s = displacement of block along wedge surface.

Equation of sloping face

$$y = -\tan \theta (x - X) + h \quad - (2)$$

Contact constraint - block must lie on wedge surface

$$\phi = y_b + \tan \theta (x_b - X) - h = 0 \quad - (3)$$

substitute (1) in (3).

$$\phi = (h - s \sin \theta) + \tan \theta (X + s \cos \theta - X) - h$$

$$\phi = -s \sin \theta + s \tan \theta \cos \theta$$

$$= \tan \theta \cdot \cos \theta = \sin \theta$$

$$\phi = s(-\sin \theta + \sin \theta) = 0$$

Constraint Eqn :- $\phi = y_b + \tan \theta (x_b - X) - h = 0$

9. The linear momentum of the system (wedge + box)

Problem 9.

For system (wedge + block) linear momentum is conserved - read as one system.

$$\frac{dP}{dt} = \sum F_{\text{external}}$$

For linear momentum to be conserved the net external force acting should be zero.

	$M_1=0, M_2=0$	$M_1>0, M_2=0$	$M_1=0, M_2>0$	$M_1>0, M_2>0$
Horizontal direction	Yes	Yes	No	No
Vertical direction	No	No	No	No

Case 1 :- $M_1 = 0, M_2 = 0$

Horizontal $\rightarrow$ Yes - Conserved

Reason - Because the M_2 is the external force acting between wedge & ground is zero & M_1 is internal friction force ~~to~~ between wedge & box which cancel out each other (Newton 3rd Law)

$$\frac{dP}{dt} = 0$$

Vertical $\rightarrow$ No - Not conserved.

Reason :- Box gravity = $m_b g$.

Wedge gravity = $m_w g$.

Ground normal = N_{ground} .

The normal force acts only on wedge ~~not~~ on box

$$N_{\text{ground}} \neq (m_b + m_w)g \neq 0.$$

Case 2 :- $M_1 > 0, M_2 = 0$.

Horizontal $\rightarrow$ Yes - Conserved.

Reason :- M_1 is the internal force which does not change linear momentum.

$M_2 = 0 \rightarrow$ external force acting between wedge & ground is zero. so linear momentum is conserved.

Vertical $\rightarrow$ No - Not conserved.

Reason $\rightarrow$ $N_{\text{ground}} = (m_b + m_w)g \neq 0$.

linear momentum in y is not conserved.

Case 3 :- $M_1 = 0, M_2 > 0$.

Horizontal $\rightarrow$ No $\rightarrow$ Not conserved

Reason :- $M_2 > 0$, the friction is external between the wedge and ground, which is greater than zero, means linear momentum in x is not conserved
 $\frac{dp}{dt} \neq 0$.

Vertical :- ~~No~~ $\rightarrow$ No - Not conserved

$$N_{\text{ground}} = (m_b + m_w)g \neq 0$$

linear momentum in y is not conserved.

Case 4 :- $M_1 > 0, M_2 > 0$.

Horizontal $\rightarrow$ No $\rightarrow$ Not conserved

Reason :- Same as in ~~Case 3~~ Case 3

$M_2 > 0$. ~~the~~ linear momentum in x not conserved

$M_1 > 0$ internal force - does not change linear momentum.

Vertical $\rightarrow$ No - Not conserved

Reason :- $N_{\text{ground}} = (m_b + m_w)g \neq 0$

linear momentum in y not conserved

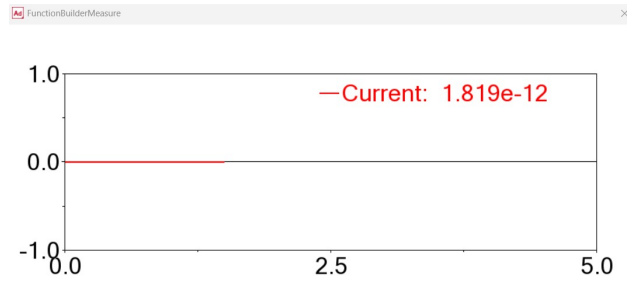


Figure 10: Linear horizontal momentum CASE:1

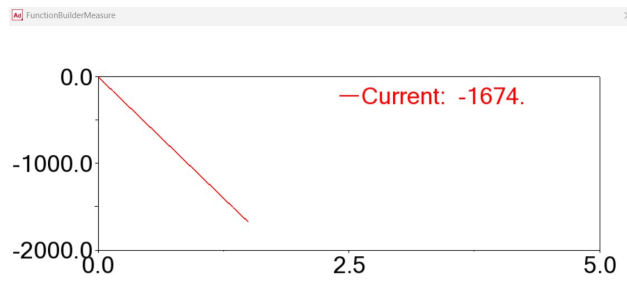


Figure 11: Linear vertical momentum CASE:1

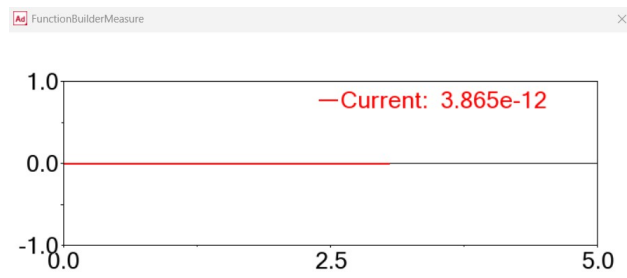


Figure 12: Linear horizontal momentum CASE:2

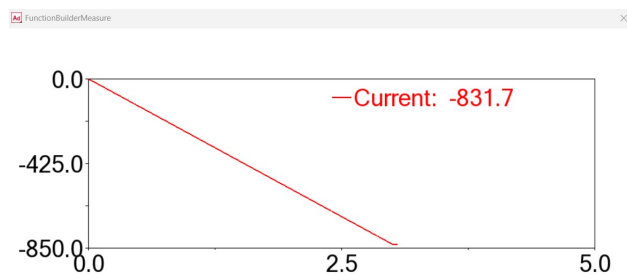


Figure 13: Linear vertical momentum CASE:2

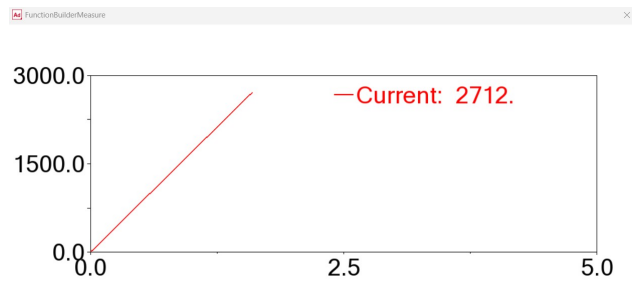


Figure 14: Linear horizontal momentum CASE:3

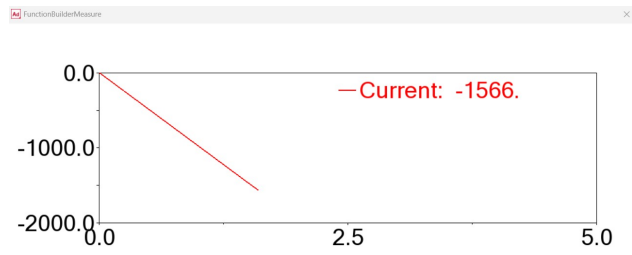


Figure 15: Linear vertical momentum CASE:2

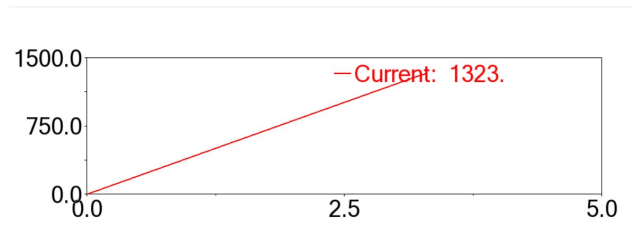


Figure 16: Linear horizontal momentum CASE:4

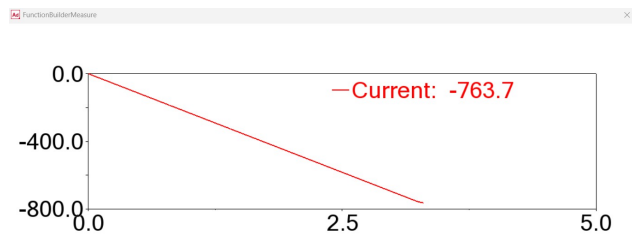
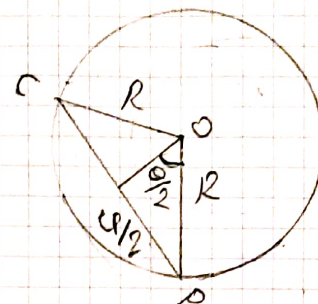
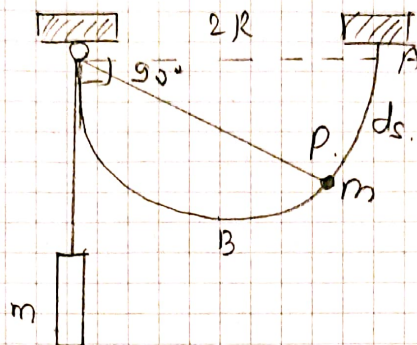


Figure 17: Linear vertical momentum CASE:4

Part 3: Problem from the course compendium.

10. Problem 1.13 in Bostrom.

Problem 10.



$$\sin \frac{\theta}{2} = \frac{CP}{2R} \Rightarrow \frac{CP}{2R}$$

$$\boxed{CP = 2R \sin \frac{\theta}{2}}$$

chord length.

R = Radius of semicircle.

m = mass of pearl = hanging mass.

CP = chord length.

θ = Angle.

$\theta = 0$ At point C.

$\theta = \frac{\pi}{2}$ At point A.

$\theta = \pi$ At point B.

The straight line chord distance from C to P on a circle of angle θ is

$$CP(\theta) = 2R \sin \frac{\theta}{2}$$

At point A.

$$CP_A = CP\left(\frac{\pi}{2}\right) = 2R \sin \frac{\pi}{4} = 2R \frac{\sqrt{2}}{2} = R\sqrt{2}$$

At point B.

$$CP_B = CP(\pi) = 2R \sin \frac{\pi}{2} = 2R.$$

Difference in chord length.

$$\Delta CP = CP_B - CP_A = 2R - \sqrt{2}R = \underline{(2 - \sqrt{2})R}$$

Total string length is constant, the increase in chord length means hanging mass must rise by same amount

V_p = speed of pearl measured along arc

V_h = speed of hanging mass.

S_p = length of arc

$$S_p = \underline{R\theta}$$

$$V_p = \frac{dS_p}{dt} \Rightarrow \frac{d(R\theta)}{dt} = R\dot{\theta}$$

$$\text{Total string length} = S_h + CP(\theta) = S_h + 2R \sin \frac{\theta}{2}$$

$$\text{differentiate } \frac{dS_h}{dt} + \frac{dCP(\theta)}{dt} \Rightarrow \dot{S}_h + 2R \cdot \frac{1}{2} \cos \frac{\theta}{2} \dot{\theta}$$

$$= \underline{\dot{S}_h + R \cos \frac{\theta}{2} \dot{\theta}}$$

$$V_h = \dot{S}_h$$

$$V_h + R \cos \frac{\theta}{2} \dot{\theta} \Rightarrow V_h + V_p \cos \frac{\theta}{2} = 0$$

$$V_h = -V_p \cos \frac{\theta}{2}$$

Pearl Pearl

When Pearl moves from A to B Potential energy

$$\Delta U_{\text{pearl}} = mg(y_B - y_A) = -mgR$$

Pearl

when Pearl moves from A to B potential energy

$$\Delta U_{\text{pearl}} = mg(y_B - y_A) = mgR$$

Hanging Mass

Hanging mass rises by ΔCP potential energy

$$\Delta U_{\text{hang}} = mg \Delta CP = mgR(2 - \sqrt{2}).$$

Total potential Energy

$$\begin{aligned}\Delta U &= \Delta U_{\text{pearl}} + \Delta U_{\text{hang}} \\ &= mgR + mgR(2 - \sqrt{2})\end{aligned}$$

$$\Delta U = mgR(3 - \sqrt{2})$$

~~ΔU~~

Total kinetic energy at B

$$K_B = \Delta U = mgR(3 - \sqrt{2})$$

$$\begin{aligned}K_B &= \frac{1}{2} m v_B^2 + \frac{1}{2} m v_h^2 \Rightarrow \frac{1}{2} m v_B^2 + \frac{1}{2} m \left(\frac{v}{\sqrt{2}} \right)^2 \\ &= \frac{3}{4} m v_B^2\end{aligned}$$

$$\frac{3}{4} m v_B^2 = mgR(3 - \sqrt{2})$$

$$\boxed{v_B = \sqrt{\frac{4}{3} g R (3 - \sqrt{2})}}$$

Velocity of pearl ~~at~~ position B.